INCH-POUND

A-A-52441
March 17, 1993
SUPERSEDING
(See 5.5)

COMMERCIAL ITEM DESCRIPTION

MOTORS, WINDSHIELD WIPER; PNEUMATIC, AND VALVES, CONTROL FOR

The General Services Administration has authorized the use of this commercial item description (CID) as a replacement for MIL-M-12185C description of type II air pressure operated windshield wiper motors and control valves. MIL-M-12185C is inactivated for acquisition of these parts. This CID also replaces MS53017A and MS53018A, which are canceled.

1.0 Abstract. This CID covers air windshield wiper motors and windshield wiper motor shut-off and speed control valve for use in military vehicles. Services and repair parts for motors and valves covered herein are not intended to be carried in the supply system. The windshield wiper motor shut-off and speed control valve is referred to herein as the control valve.

1.1 Classification. Windshield wiper motors shall be of the following classes:

Class I - Pneumatic, universal
Class II - Pneumatic

2.0 Salient characteristics.

2.1 Materials. Unless otherwise specified herein, the materials used shall be in accordance with the manufacturer's material specifications for windshield wiper motors and control valves. The use of recovered material made in conformance with regulatory requirements is acceptable providing that all requirements of this CID are met (see 5.6).

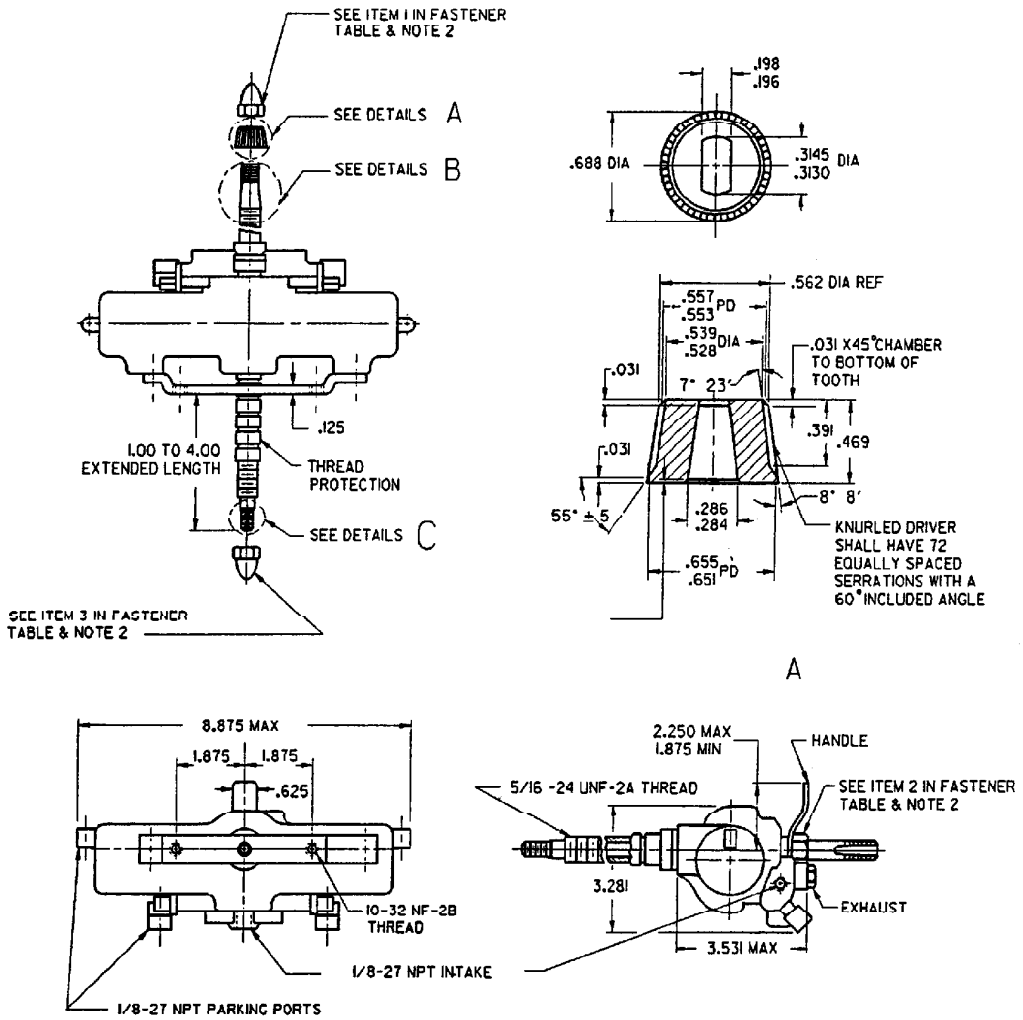
2.2 Design and construction. Unless otherwise specified herein, windshield wiper motors and control valves shall be designed and constructed in accordance with the manufacturer's specifications/drawings. Class I and II windshield wiper motors shall be physically and functionally interchangeable.

Beneficial comments, recommendations, additions, deletions clarifications, etc. and any other data which may improve this document should be sent by letter to: U.S. Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000.

AMSC N/A FSC 2540

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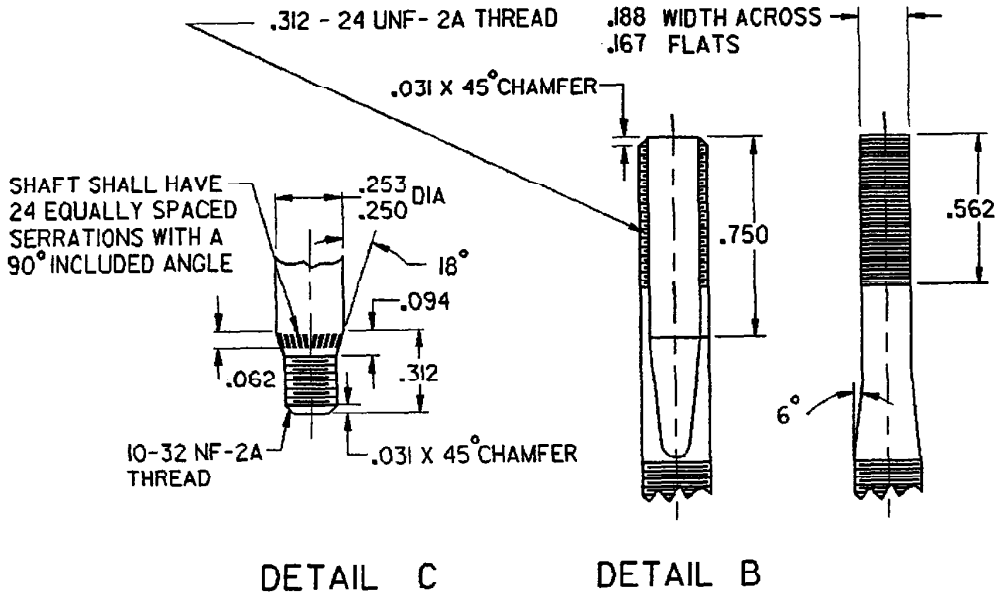


FASTENER TABLE			
ITEM NO.	NAME	SIZE	REPLACEMENT PART NO.
1	NUT, PLAIN, CAP HIGH CROWN	.3125-24 UNF-2B	MS24680-66
2	NUT, PLAIN, HEXAGON	.3125-24 UNF-2B	MS51968-5
3	NUT, PLAIN, CAP HIGH CROWN	.190-32 UNF-2B	MS24680-63

CROSS REFERENCE		
CID PART NO.	FORMER MS PART NO.	FORMER ARMY PART NO.
A52441-1	MS53017-1	8379907

FIGURE 1. Motor, windshield wiper, pneumatic, universal.

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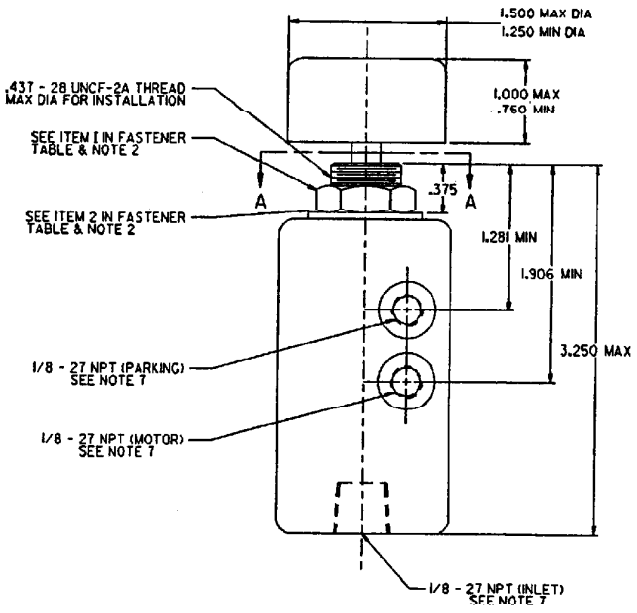


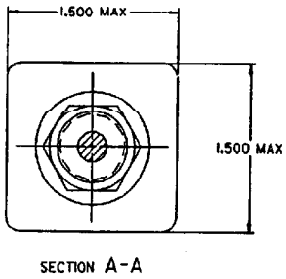
NOTES:

1. Material of knurled driver (Detail "A") shall be zinc plated sintered iron. Before plating, knurled driver shall resist 1700 lbs. compression without breaking while mounted on motor drive shaft.
2. Material of cap nuts and plain nut shall be zinc plated carbon steel.
3. Shaft thread protector shall be indented in increments of 1/2 inch and of such material as to allow cutting at indentations to obtain required length of assembly. When necessary, protector shall be cut at the unthreaded end.
4. Parking, inlet, and exhaust ports shall be identified by stamping, casting or molding.
5. Location of parking, intake, and exhaust ports optional; however, ports shall provide optional access for connection when motor is in position.
6. Dimensions are in inches. Unless otherwise specified, tolerances are $\pm$ .015 inch on decimals and $\pm$ 1° on angular dimensions.
7. This CID is not intended to limit construction to features other than as shown herein by dimensions or notations.

FIGURE 1. Motor, windshield wiper, pneumatic, universal - Continued.

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SECTION A-A

FASTENER TABLE			
ITEM NO.	NAME	SIZE	REPLACEMENT PART NO.
1	NUT,PLAIN, HEXAGON	.4375-20 UNF-2B	MS51968-11
2	WASHER,FLAT, ROUND	.531 X 1.062 X .095	MS15795-818

CROSS REFERENCE	
CID PART NO.	FORMER MS PART NO.
A5244I-V	MS53018-1

NOTES:

1. Material of valve body shall be aluminum, brass or zinc alloy die casting.
2. Material of washer shall be corrosion resistant steel. Material of nut shall be zinc plated carbon steel.
3. Construction and material of control knob is optional, but shall be adequate for intended use.
4. Control knob shall be inscribed to specify its use and shall indicate direction of operation.
5. Construction of body optional (round, ellipsoidal, etc.) within dimensions given.
6. Location of inlet port optional in end of body indicated.
7. Parking, motor, and inlet ports shall be indicated by stamping, casting, or molding.
8. Dimensions are in inches. Unless otherwise indicated tolerances are ± 0.15 inch on decimals.
9. This CID is not intended to limit construction to features other than shown herein by dimensions or notations.

FIGURE 2. Control valve.

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2.2.1 Manual operation. When specified in applicable drawings, windshield wiper motors shall be equipped with a handle or lever to permit manual operation. The design shall permit an operator wearing heavy gloves or mittens to operate the motor.

2.2.2 Adjustable cleaning arc. The cleaning arc of class I windshield wiper motors shall be adjustable within a minimum range of 40, 100 and 140 degrees (°). The arc setting shall remain fixed within ± 3° after operating at 90 pounds per square inch (psi) for 30 minutes at each arc setting. Arc setting shall be made without disassembly of the wiper motor and shall be capable of being adjusted when mounted for intended operation.

2.2.3 Parking device. Class I windshield wiper motors shall incorporate, in combination with the control valve a provision for moving the wiper blade to an extreme position when the control is turned to "off" position. Class II motors shall include such provisions when specified in applicable drawings.

2.2.3.1 Manual parking override. Class I windshield wiper motors shall have a right and left manual parking override of at least 30 degrees beyond the maximum cleaning arc.

2.2.4 Adjustable shaft. Class I windshield wiper motors shall be designed to incorporate a reversible drive shaft capable of being adjusted within a range from 1 to 4 inches (see figure 1). Reversal of the drive shaft shall permit either the large or small drive end of the shaft to be extended from the driving end of the wiper motor. Shaft adjustment or reversal shall be accomplished without disassembly of the wiper motor.

2.3 Performance.

2.3.1 Torque, speed and air consumption. Windshield wiper motors shall meet the requirements shown in table I:

TABLE I. Ambient performance.

Air pressure, psi	Torque, lb-in. minimum (min)	Speed, cpm (min)	Air consumption cfm maximum (max)
20	12	85	0.35
30	20	95	0.50
50	40	100	0.70
80*	65*	105*	1.35*
90*	70*	105*	1.35*

*Applicable to class I (universal) motor only

2.3.2 Control valve. With an air pressure of 90 psi for Class I motors and 50 psi for Class II motors, the control valve shall regulate pneumatic wiper motor speed smoothly and uniformly over a speed range from 30 to 105 cycles per minute (cpm).

2.3.3 High temperature. When operated at ambient temperature of 160 ± 5 degrees Fahrenheit (°F), following a 5 hour heat soak at the specified temperature, windshield wiper motors shall meet the requirements shown in table II.

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TABLE II. High temperature performance.

Air pressure, psi	Torque, lb-in. (min)	Speed, cpm (min)	Air consumption cfm (max)
20	12	90	0.30
30	20	100	0.45
50	40	105	0.65
80*	60*	110*	1.00*
90*	75*	110*	1.20*

*Applicable to class I (universal) motor only

2.3.4 Low temperature. When operated at ambient temperature of minus 20 ± 5°F, following a 5 hour cold soak at the specified temperature, windshield wiper motors shall meet the requirements shown in table III:

TABLE III. Low temperature performance.

Air pressure, psi	Torque, lb-in. (min)	Speed, cpm (min)	Air consumption cfm (max)
20	12	40	0.55*
30	20	60	0.80*
50	37	75	1.30*
80*	60*	85*	1.75*
90*	68*	95*	1.85*

*Applicable to class I (universal) motor only

2.4 Protective finish. Exterior surfaces of motors and valves, except pipes or screw threads, shall be painted or dyed olive drab. Motors and valves shall meet the requirements of 2.3.1 and 2.3.2 after 200 hours exposure to a 5% salt spray at 95°F for 200 hours. The 5% salt solution shall have a ph of 6.5 to 7.2. Following the salt spray, the motors and valves shall not show any evidence of bare base material, undercoats, flaking paint or whitening.

2.4.1 Metal parts. The exposed metal surfaces of the motors and valves shall be corrosion resistant and shall meet the requirements of 2.3.1 and 2.3.2 after 200 hours exposure to the salt spray.

2.4.2 Nonmetallic parts. All nonmetallic parts shall be treated to resist the effects of salt spray. After treatment they shall evidence no functional failure when subjected to 200 hours exposure to the salt spray.

2.5 Identification marking. Identification marking shall be permanent and legible and shall include, as a minimum, the part identification number (PIN), the manufacturer's identification code (CAGE) and the national stock number (NSN).

2.6 Instructions. Illustrated instructions for installation, operation, setting and adjustment shall be packaged with each windshield wiper motor and control valve.

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3.0 Quality assurance provisions.

3.1 Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

3.2 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description and that the product conforms to the contractor's own drawings, specifications, workmanship, standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.0 Preservation, packaging, packing, labeling, and marking. Preservation, packaging, packing, labeling, and marking for the desired level shall be as specified in the contract (see 5.2).

5.0 Notes.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

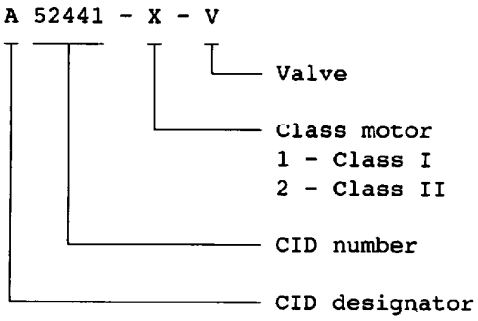
5.1 Addresses for obtaining copies of referenced documents.

5.1.1 Non-Government publications. ASTM E380-91A "Standard Practices for Use of the International System of Units (SI)" (the Modernized Metric System) is available from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.

5.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this CID.
b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 5.1).
c. Class of windshield wiper motor.
d. Applicable drawing if required (see 2.2).
e. Selection of applicable level and packaging requirements (see 4.0).

5.3 Part or identification number (PIN). The PINs to be used for windshield wiper motors and valves acquired to this CID are created as follows:



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5.4 Metric products. Motors and valves that are to metric dimensions will be considered on the following basis:

a. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of ASTM E380, and all other requirements of this CID are met.

b. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

5.5 Supersession and cross reference data. The CID supersedes MIL-M-12185C, dated 23 October 1987, MS53017A, dated 9 July 1968, and MS53018A, dated 9 July 1968. Windshield wiper motors and control valves conforming to this CID are interchangeable/substitutable with windshield wiper motors and valves conforming to the superseded documents.

5.6 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

MILITARY INTERESTS:

Custodians
Army - AT
Navy - YD

User activities
Navy - MC
DLA - CS

Review activity
Army - CR

CIVIL AGENCY COORDINATING ACTIVITY:
GSA - FSS

PREPARING ACTIVITY:
Army - AT

(Project 2540-0380)

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